Except, they weren't.

After just a few months, physicists and astronomers outside the BICEP2 collaboration independently analyzed the data and found that the pattern could be completely explained by something much more mundane: ordinary cosmic dust, in our own Milky Way galaxy. If primordial gravitational waves *had* been discovered, they would have been evidence against the ekpyrotic model, since that model doesn't include the inflationary universe-quake that could produce them. Unfortunately, their nondetection takes us back to square one. While inflation theory says that primordial gravitational waves must be produced, there's nothing in the theory that says they have to be *detectable*. The most popular inflation models give you substantial gravitational waves, but it's entirely possible to come up with one that produces a signal too weak to compete with the confusion of the cosmic dust.[fn8] So the fact that the dust got in our way doesn't prove that the inflation signal isn't there, any more than it proves that it is.

Still, we might get clues from other sources. We might find evidence for or against braneworlds in the search for extra dimensions, or we might finally get a hint of those primordial gravitational waves. Even ordinary gravitational waves could hold clues, either by showing us a signal that travels through the bulk (via interdimensional aliens or not),[fn9] or by helping us to map out the structure of spacetime by, essentially, watching how it wiggles. According to some studies, data from black hole collisions have already put a damper on theories involving gravity leaking out into a higher-dimensional void. So far, all our measurements are consistent with a plain old boring universe with only three spatial dimensions.

Whether or not we find extra dimensions, the idea of a cyclic universe will likely continue to hold appeal as an alternative to inflation. One reason is the problem of entropy, the ever-increasing disorder in the universe that ultimately leads to a Heat Death. We can calculate the amount of entropy in our observable universe, and we can look back through cosmic history to determine what it must have been at early times if it has been steadily increasing over the lifetime of the cosmos. The result is that the universe must have started at a shockingly low-entropy—highly ordered—state when our own cosmic history began. This is a deeply uncomfortable idea for a lot of cosmologists. How did the entropy get set so low at the beginning? It's as if you walk into a room you're sure no one has ever been in before and you find rows and rows of dominoes lying on the floor,